

Tony Mogoao Ombaso

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Education

University of Pretoria

MSc Computer Science

Pretoria, South Africa

Feb 2026 - Present

- Masters by research at the Computational Intelligence Research Group
- Working title: Loss Landscape Analysis of the Lottery Ticket Hypothesis

University of Pretoria

BScHons Computer Science

Pretoria, South Africa

Feb 2025 - Nov 2025

- With distinction, Grade: 82.38%
- Coursework: Computer Science Education; Scientific Research; Artificial Intelligence I and II; Formal Aspects of Computing (Model Checking); Natural Language Processing; Image Processing

Strathmore University

BSc in Informatics and Computer Science

Nairobi, Kenya

July 2019 - July 2023

- First Class Honours, Grade: A (84.14%)
- Graduated top of the class, the best student in the School of Engineering and Computing Sciences.
- Coursework: Data Structures and Algorithms; Programming; Software Engineering, Web Application Programming; API development; Scientific Research; Artificial Intelligence; Machine Learning; Software Engineering; Databases; Cryptography and CyberSecurity

Research Experience

Honours Research Project (mark: 94/100)

University of Pretoria

Pretoria, South Africa

Feb 2025 - November 2025

- Research report titled: A Python Library for Explainable Problem Landscape Analysis (working towards publishing).
- Developed and published a Python library for Explainable Landscape Analysis (xLA) comprising 17 xLA features (link to PyPI package).
- Showed that the developed library is effective for xLA.
- **Technical Skills:** PyTorch, Google Colab, Python

Natural Language Processing module

University of Pretoria

Pretoria, South Africa

Feb 2025 - June 2025

- Semester research report titled: Meta-learning for Semantic Relatedness: A Comparative Study of Meta-Learning and Transfer Learning for Semantic Relatedness Tasks in African Languages using the SemRel2024 Dataset.
- Aim: determine whether meta-learning (model-agnostic meta-learning) could outperform transfer learning in low-resourced African languages STR tasks (link to repo).
- Showed that transfer learning outperformed the meta-learning approach.
- **Technical Skills:** PyTorch, Google Colab, Python

Undergraduate Final Research Project (mark: 81.94)

Strathmore University

Nairobi, Kenya

Apr 2022 - Dec 2022

- Developed a deep learning model for segmentation and classification of melanoma skin cancer lesions.
- Carried out a comparative study of fine-tuning 6 neural network architectures: ResNet34, ResNet50, ResNet152, InceptionV3, VisionTransformer, and EfficientNet-B7.
- Trained on an ensemble dataset of 5 different medical image datasets: MED-NODE, PAD-UFES-20, 7-point, DermIS, and DermQuest.
- Developed mobile app for taking and uploading photos of skin lesions to a RESTful API for inference.
- Sep 2024: Recognised as a 2024 Highly Commended Entrant and subsequently as the 2024 Africa and the Middle East Regional Winner by the Global Undergraduate Awards.
- Published on The Global Undergraduate Award Library (HERE)
- **Technical Skills:** PyTorch, Flask, SQLite, React Native, Google Colab, Python, Talend

Software Engineering Experience

Strathmore University @iLabAfrica

Junior Software Developer

Nairobi, Kenya

Jan 2023 - Jan 2025

- Collaborating with a four-person team in the development of the Public Procurement Information Portal (PPIP). A portal, mandated by law and used country-wide for the accountability of the use of public funds.
- Migrated data from the old PPIP portal to the new enhanced one.
- Implemented the Open Contracting Data Standard (OCDS) on the PPIP portal to comply with international standards.
- Deployed the PPIP portal on a production environment (link).
- **Technical Skills:** Laravel (PHP), Vue.js (JavaScript), Cypress, MySQL, Docker, Test Driven Development, Linux, Git, GitLab

Cafeteria ChapChap

Co-founder

Nairobi, Kenya

Oct 2021 - Mar 2022

- Developing a mobile application that allows users at Strathmore University to order and pay for meals from their phone.
- The goal is to significantly improve queueing time by eliminating dependence on human cashiers.
- The project won **1st prize** in the first Strathmore University's Ideas Festival.
- **Skills:** LDAP, React Native, Laravel, Docker, Typescript, Test Driven Development (TDD), Project Management

Skills

Programming	Systems Programming: Laravel, MySQL, Vue.js, React.js, Typescript, Git, Docker, Test Driven Development, Linux
Artificial Intelligence	Python, PyTorch, Deep Learning, Jupyter Notebook
Scientific Research	LaTeX, Research Writing
Miscellaneous	UI/UX Design, Poster Design

Volunteering

AI Literacy Society

University of Pretoria, South Africa

Chairperson

Jan 2026 - Present

- Leading registration and organising activities towards improving AI literacy within the University of Pretoria community.

Strathmore University

Nairobi, Kenya

Mentor

Sep 2023 - Dec 2025

- Mentoring 5 undergraduate students pursuing undergraduate degrees in Computer Science.
- Mentoring involves informal meetings with each mentee one-on-one to accompany them with decisions in their life.

Mbagathi Study Center

Nairobi, Kenya

Secretary

Jan 2024 - Present

- Coordinating the mentorship activities for university students:
- Organised 3 weekend-long character seminars
- Organised 9 charity visits to charitable organisations attending to vulnerable groups
- Organised 1 conference for university students on the impact of AI on our humanity.
- Mentoring 4 students of Lenana High School.
- Giving class on virtuous leadership

Achievements

2024	Global Undergraduate Awards Regional Winner , Africa & Middle East of the Computer Science Category	Dublin, Ireland
2024	Global Undergraduate Awards Highly Commended , in the Computer Science Category	Dublin, Ireland
2023	Academic Award , Best undergraduate in the School of Computing and Engineering Sciences.	Strathmore University
2022	Ideas Festival Winner , Won 1st prize in an innovation competition	Strathmore University
2022	Dean's List Award , Best student overall in my 3rd year of undergraduate.	Strathmore University
2021	Dean's List Award , Best student overall in my 2nd year of undergraduate.	Strathmore University
2020	Dean's List Award , Best student overall in my 1st year of undergraduate.	Strathmore University

Interests and Hobbies

Leisure Reading

I enjoy reading contemporary novels, history, biographies, classics.

Philosophy

Metaphysics, Philosophy of Language, Philosophy of Science, Philosophy of Nature, Aesthetics